TECHNICAL REPORT • PILOT STUDY • COMPUTER VISION FOR AGRICULTURE

When Labels Are the Bottleneck

A Pilot Study of Active Learning for Closing the Lab-to-Field Gap in Plant Disease Recognition

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Abstract. Convolutional networks trained on laboratory photographs of diseased leaves routinely report accuracies above 99%, yet the leaves a farmer actually photographs look nothing like the laboratory ones. We first measure this gap, then ask whether a small labeling budget can narrow it. A ResNet-18 fine-tuned on 28,220 PLANTVILLAGE images reaches 99.43% validation accuracy but only 25.44% accuracy (macro-F_1 0.213) on 228 held-out PLANTDOC field images drawn from the same 26 classes, a drop of 74 points. Because field images need expert labels (PLANTDOC reports about 300 human-hours for roughly 2,600 images), we treat annotation as the scarce resource and compare three ways of spending a budget of 100 queries on top of 130 seed labels: random, entropy-based uncertainty, and feature-space diversity sampling. Over three seeds, uncertainty sampling attains the highest mean macro-F_1 (0.333 ± 0.028, against 0.309 ± 0.062 for random and 0.298 ± 0.027 for diversity), but the ranking changes with the seed, and the differences are comparable to or smaller than the sampling noise of a 228-image test set. The robust finding is a different one: 230 in-domain labels, 0.8% of the laboratory data, outperform the laboratory-trained model by about 8 accuracy points. We present the work as a pilot, state its threats to validity, and outline the protocol that would be needed to resolve the question it leaves open.

Keywords: active learning · plant disease classification · domain shift · uncertainty sampling · diversity sampling · PlantVillage · PlantDoc

Key findings at a glance

- **The gap is enormous.** 99.43% accuracy on laboratory images becomes 25.44% on field images of the same 26 classes (chance is 3.8%).
- **A few field labels beat a mountain of lab labels.** Models trained on only 230 field images average 33.1% accuracy over nine runs, versus 25.4% for the model trained on 28,220 lab images.
- **Choosing *which* images to label is second-order at this scale.** Uncertainty sampling had the best mean macro-F_1 and random sampling twice the seed-to-seed spread, but the winner changed from seed to seed.
- **Same data, different outcome.** The identical 230 images produced 32.0% accuracy in one run and 39.9% in another, a 7.9-point swing that is larger than any gap between strategies.

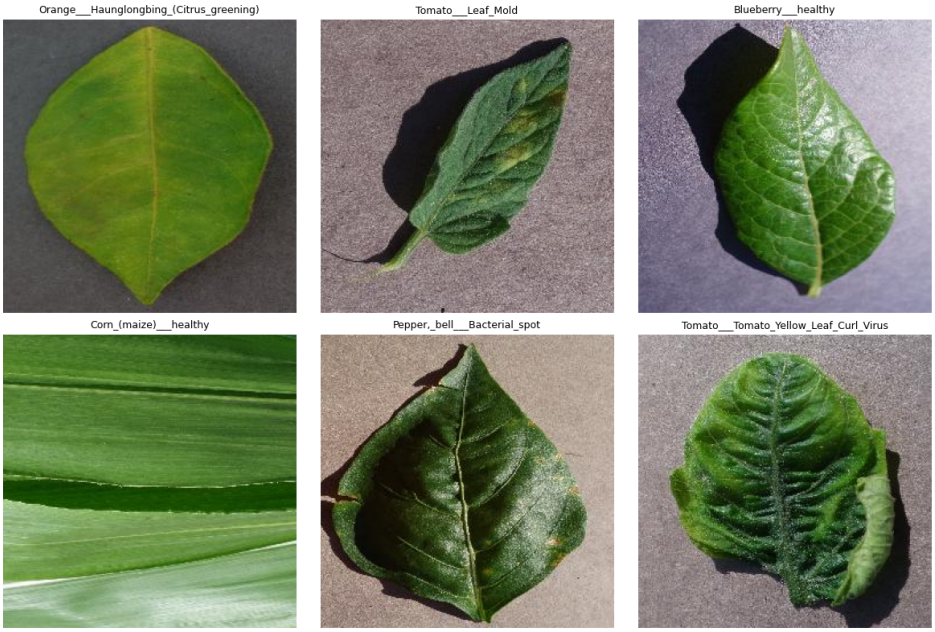


Figure 1. Six random images from the PLANTVILLAGE color set (all 38 classes). Each shows a single leaf photographed against a plain, uniform background: the visual regime in which published accuracies above 99% are obtained. Images produced by the project's own notebook.

1 Introduction

1.1 The 99% illusion

Image-based plant disease recognition is one of the most celebrated success stories of applied deep learning. Since the release of the PLANTVILLAGE repository [7] and the first deep-learning benchmark on it [10], accuracies above 99% on held-out test images have become routine, and an ImageNet-pretrained ResNet-18 [4, 6] needs only a few epochs to reach them. Taken at face value, plant disease diagnosis looks solved.

It is not. The images in PLANTVILLAGE were photographed under controlled conditions: one leaf, detached, laid flat on a uniform gray or black background, evenly lit (Figure 1). The images that matter in practice are taken by a smallholder with a phone, in a field, with soil, hands, shadows, other leaves, and a leaf that is only partly in frame. A classifier that has only ever seen the first kind of picture has no reason to work on the second, and, as we will show, it largely does not.

1.2 The real purpose of this study

The question behind this work is not "which architecture tops the leaderboard?" but rather an economic one:

> *When the images that matter are the ones that are hardest and most expensive to label, which images should an expert be asked to label first?*

The PLANTDOC dataset [17] was built precisely because lab-trained models fail in the field. Its authors report that assembling 2,598 annotated images across 13 plant species and up to 17 disease classes took approximately 300 human-hours of effort, and they motivate the work by noting that plant diseases cost India about 35% of its annual crop yield. Annotation by someone able to tell early blight from Septoria leaf spot is therefore the binding constraint, not GPU time. Under such a constraint the central design question of *active learning* [15] becomes practical: given a pool of unlabeled field photographs and a budget of b expert labels, can a model's own uncertainty, or the geometry of its feature space, pick a

better b images than a coin flip?

This report is a small, carefully controlled attempt to answer that question, and it is deliberately framed as a *pilot*. We want to know, concretely:

RQ1. How large is the lab-to-field gap on a harmonized, leakage-checked benchmark?

RQ2. How much of the gap can be recovered with a very small number of in-domain labels?

RQ3. Does uncertainty sampling or diversity sampling beat random sampling at that budget, and is the answer stable across seeds?

1.3 Contributions

- A harmonized 26-class benchmark linking PLANTVILLAGE and PLANTDOC, with an exact-duplicate audit (SHA-256) across the two corpora (Section 4).
- A quantification of the lab-to-field gap and a per-class diagnosis of how the lab-trained model fails (Section 6.1).
- A controlled comparison of random, uncertainty, and diversity sampling under identical seed sets, budget, architecture, and training recipe, repeated over three seeds (Section 6.3).
- An honest account of *ranking instability*: how the apparent winner changes between an exploratory run, a controlled run, and a multi-seed average, together with an argument for why three seeds cannot settle the question (Sections 6.5 and 6.7).

2 Background and Related Work

Lab versus field datasets. PLANTVILLAGE [7] contains tens of thousands of leaf images in 38 crop–disease classes photographed on plain backgrounds; Mohanty et al. [10] showed that deep networks reach very high accuracy on it. Subsequent analysis has cautioned that such accuracy can be partly driven by dataset artifacts such as background statistics [11]. PLANTDOC [17] instead scrapes images from the internet and annotates them by hand; they are visually heterogeneous, and the authors show that training on them improves field classification accuracy by up to 31% compared with relying on lab data alone. Complementary work applies YOLOv8n-seg to plant-disease detection and instance segmentation, addressing localization rather than the image-level classification considered here [1].

Domain shift. Training and deployment distributions differ in the *covariate shift* sense: $p_{\text{lab}}(x) \neq p_{\text{field}}(x)$. Learning-theoretic bounds on target error grow with the divergence between source and target distributions [2], which is the formal version of the intuition that more lab images cannot, on their own, fix a field problem.

Active learning. Active learning [15] replaces passive random annotation with an interactive loop in which the learner chooses the examples to be labeled. Two families matter here. *Uncertainty sampling* [8] queries the examples about which the current model is least certain; with predictive entropy [16] as the uncertainty score it is the textbook baseline. *Diversity* or *core-set* methods [14] instead choose a batch that covers the feature space, so that every unlabeled example is close to some labeled one. A well-known caveat is the *cold start*: with very few labels the model's uncertainty and features are unreliable, and recent work argues that the best strategy at very low budgets can be the opposite of what works at high budgets [5]. Our budget of 100 queries on top of 130 seed labels sits squarely in that regime, which is one reason we treat the outcome as an open empirical question rather than a foregone conclusion.

Table 1. Entropy as an uncertainty score, for five unlabeled PLANTDOC images scored by an intermediate model during development. Higher entropy means a less certain prediction; the maximum possible value is $\ln 26 \approx 3.258$.

Image	Predicted class	Top probability	Entropy H (nats)
1	Tomato early blight	0.248	**2.583**
2	Tomato yellow leaf curl virus	0.629	1.572
3	Peach healthy	0.924	0.461
4	Bell pepper bacterial spot	0.701	1.381
5	Tomato late blight	0.896	0.633

3 Problem Formulation

Let $\mathcal{C}$ be a set of $C = 26$ crop–disease classes. We have a labeled *source* set $\mathcal{S}$ of laboratory images, a *target pool* $\mathcal{P}$ of field images whose labels are hidden until queried, and a held-out *target test set* $\mathcal{T}$. The learner starts with a small labeled seed $\mathcal{L}_0 \subset \mathcal{P}$ and may query an oracle for the labels of b further pool images $\mathcal{Q} \subset \mathcal{P} \setminus \mathcal{L}_0$. After labeling, a classifier $f_\theta : \mathcal{X} \to \Delta^{C-1}$ is trained on $\mathcal{L} = \mathcal{L}_0 \cup \mathcal{Q}$ and evaluated on $\mathcal{T}$. The goal of a query strategy σ is

$$\mathcal{Q}^\star_\sigma \;=\; \sigma\big(f_{\theta_0},\, \mathcal{P} \setminus \mathcal{L}_0,\, b\big), \qquad \text{maximizing} \quad \mathbb{E}\big[\text{macro-}F_1\big(f_{\theta(\mathcal{L}_0 \cup \mathcal{Q})};\, \mathcal{T}\big)\big], \tag{1}$$

where f_{θ_0} is a selection model trained on the seed set only.

Metrics. We report accuracy $\text{Acc} = \frac{1}{|\mathcal{T}|}\sum_i \mathbf{1}[\hat{y}_i = y_i]$ and macro-averaged F_1,

$$\text{macro-}F_1 \;=\; \frac{1}{C}\sum_{c \in \mathcal{C}} \frac{2\,P_c R_c}{P_c + R_c}, \tag{2}$$

with the convention that a class with no correct predictions contributes zero. Macro-F_1 weights every class equally and therefore penalizes models that ignore rare or difficult classes, which is the failure mode that matters for disease diagnosis; accuracy is reported alongside it for comparability with the literature.

Query strategies. We study three, all in the *batch* setting (one round, $b = 100$ queries selected at once).

Random. $\mathcal{Q}$ is drawn uniformly from the unlabeled pool.

Uncertainty (entropy). For each unlabeled x with predicted class distribution $p(x) = f_{\theta_0}(x)$,

$$H(x) \;=\; -\sum_{c=1}^{C} p_c(x)\, \ln p_c(x), \tag{3}$$

and $\mathcal{Q}$ is the set of the b pool images with the highest H. The maximum possible value is $\ln 26 \approx 3.258$ nats, attained by a uniform prediction. Table 1 shows how the score behaves on five pool images scored by an intermediate model during development.

Diversity (greedy farthest-point). Let $\phi(x) \in \mathbb{R}^{512}$ be the penultimate-layer (global-pooled) ResNet-18 feature of x and $s(x,z) = \frac{\phi(x)^\top \phi(z)}{\|\phi(x)\|\,\|\phi(z)\|}$ the cosine similarity. Each candidate's *coverage score* is its similarity to the nearest labeled example,

$$m(x) \;=\; \max_{z \in \mathcal{L}} s(x,z), \tag{4}$$

and the strategy repeatedly selects $x^\star = \arg\min_x m(x)$, adds it to $\mathcal{L}$, and updates every candidate's score by

$$m(x) \;\leftarrow\; \max\{\, m(x),\; s(x,x^\star)\,\}. \tag{5}$$

This is the greedy k-center rule of core-set selection [14] under cosine similarity.

Table 2. Benchmark after class harmonization (26 shared classes).

Corpus	Split	Images	Classes	Role
PLANTVILLAGE (lab)	source-train	28,220	26	trains the baseline
PLANTVILLAGE (lab)	source-validation	7,055	26	in-domain (lab) check
PLANTDOC (field)	train pool	2,285	26	active-learning pool (130 seed + 2,155 unlabeled)
PLANTDOC (field)	test	228	26	fixed held-out field test

4 Data

4.1 Source: PlantVillage

We use the original color images of the PLANTVILLAGE repository (`raw/color`), which contains 54,305 images in 38 classes. Class sizes are very uneven: the two largest of the 26 classes we keep are tomato yellow leaf curl virus (5,357 images) and healthy soybean (5,090), whereas the smallest is apple cedar rust (275).

4.2 Target: PlantDoc

We use the cropped-image classification release of PLANTDOC, which ships with an official split: 2,342 training images in 28 folders and 236 test images in 27 folders. Folder names are free-form ("Apple rust leaf", "Tomato mold leaf") and do not coincide with PLANTVILLAGE labels.

4.3 Class harmonization

We map each PLANTDOC folder to its natural PLANTVILLAGE counterpart by hand (Appendix A), giving 26 shared classes. Two PLANTDOC folders were left out because they could not be mapped confidently: the generic "Tomato leaf" folder (55 training / 8 test images) and a two-image spider-mite folder. After harmonization the benchmark has the composition in Table 2. The PLANTVILLAGE images are split 80/20 with stratification (seed 42) into a source-training and a source-validation set; the official PLANTDOC split is left untouched, so the 228 test images are never seen by any model, selection routine, or hyper-parameter decision.

4.4 Leakage audit

Because both corpora originate from public web and research sources, near-identical images could in principle appear in both. We computed SHA-256 digests of every image in PLANTDOC and of every PLANTVILLAGE color image and looked for matches: **zero** exact cross-corpus duplicates were found. This audit detects only byte-identical files; resized, re-compressed, or cropped copies would pass undetected, a limitation we return to in Section 8.

5 Experimental Protocol

The study has two stages (Figure 2).

5.1 Stage A: lab-trained baseline

An ImageNet-pretrained ResNet-18 [6] has its 1000-way head replaced by a 26-way linear layer and is fine-tuned end-to-end on the 28,220 source-training images for three epochs. It is evaluated on the 7,055 source-validation images and, once, on the 228 PLANTDOC test images.

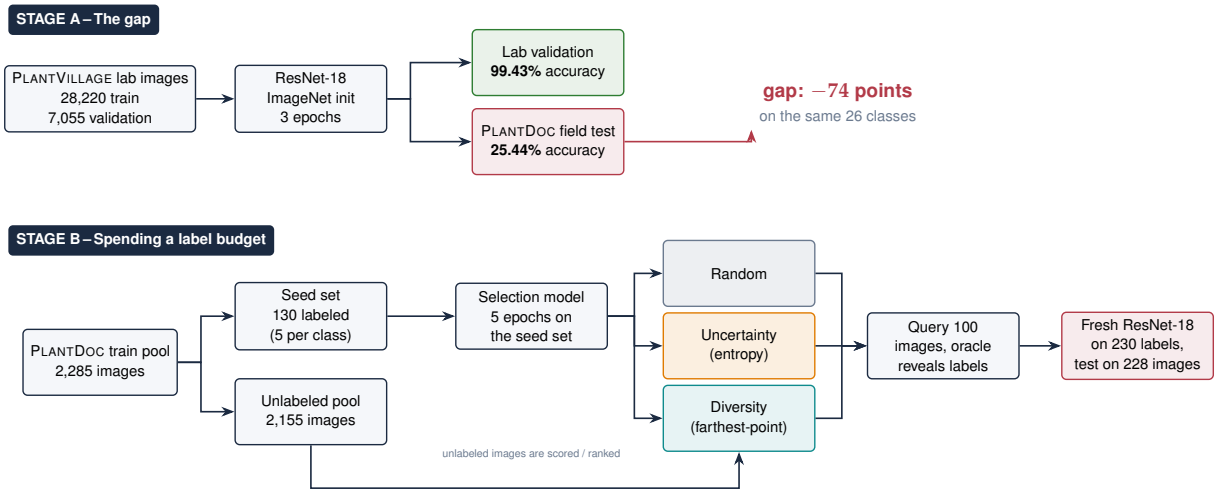


Figure 2. Study design. **Stage A** measures how a lab-trained network transfers to field images. **Stage B** treats annotation as the scarce resource: starting from 5 labels per field class, three strategies each choose 100 more images to label, and a freshly initialized network is trained on the resulting 230 labels.

Algorithm 1: One controlled active-learning run for strategy σ and seed s

Input: pool $\mathcal{P}$, test set $\mathcal{T}$, budget $b = 100$, strategy $\sigma \in \{\text{RANDOM}, \text{UNCERTAINTY}, \text{DIVERSITY}\}$

$\mathcal{L}_0 \leftarrow$ 5 images per class from $\mathcal{P}$ (seed s); $\quad \mathcal{U} \leftarrow \mathcal{P} \setminus \mathcal{L}_0$;

$\theta_0 \leftarrow$ fine-tune ImageNet ResNet-18 on $\mathcal{L}_0$ (5 epochs);

switch σ **do**

  case RANDOM **do** $\mathcal{Q} \leftarrow b$ images drawn uniformly from $\mathcal{U}$ (seed s);

  case UNCERTAINTY **do** $\mathcal{Q} \leftarrow$ the b images of $\mathcal{U}$ with largest $H(x)$ under f_{θ_0};

  case DIVERSITY **do**

    $m(x) \leftarrow \max_{z \in \mathcal{L}_0} s(\phi(x), \phi(z))$ for all $x \in \mathcal{U}$; $\quad \mathcal{Q} \leftarrow \emptyset$;

    repeat

      $x^\star \leftarrow \arg\min_{x \in \mathcal{U} \setminus \mathcal{Q}} m(x)$; $\quad \mathcal{Q} \leftarrow \mathcal{Q} \cup \{x^\star\}$;

      $m(x) \leftarrow \max\{m(x),\, s(\phi(x), \phi(x^\star))\}$ for all x;

    until $|\mathcal{Q}| = b$;

$\mathcal{L} \leftarrow \mathcal{L}_0 \cup \mathcal{Q}$ (the oracle reveals the labels of $\mathcal{Q}$);

$\theta \leftarrow$ fine-tune a *fresh* ImageNet ResNet-18 on $\mathcal{L}$ (5 epochs);

Output: Acc and macro-F_1 of f_θ on $\mathcal{T}$

5.2 Stage B: active learning on the field pool

The PLANTDOC training pool (2,285 images) is split into a *seed set* of 5 images per class (130 images, sampled with the run's seed) and an *unlabeled pool* of the remaining 2,155. Algorithm 1 gives one run. Four design decisions deserve emphasis because they make the comparison fair:

- **Identical starting point.** For a given seed, all three strategies share the same seed set, the same unlabeled pool, and the same selection model. They differ only in how the 100 queries are chosen.
- **Identical budget.** Every strategy yields exactly 230 labeled images covering all 26 classes (verified programmatically).
- **Fresh final models.** Each final classifier is re-initialized from ImageNet weights and trained from scratch on its 230 labels, so no strategy inherits training from another.
- **Deterministic training.** Python, NumPy, and PyTorch seeds are fixed, cuDNN is set to deterministic mode, and data loading is single-process.

Table 3. Training configuration. Software: PyTorch [12] 2.11.0 and torchvision 0.26.0 on a single NVIDIA Tesla T4; metrics via scikit-learn [13].

Setting	Stage A (baseline)	Stage B (active learning)
Architecture	ResNet-18, ImageNet init, 26-way head	same
Optimizer	AdamW, lr 10^{-4}, weight decay 10^{-4}	same
Loss	cross-entropy	same
Epochs	3	5
Batch size	32 (train), 64 (validation)	16
Augmentation	flip, rotation $\pm 10^{\circ}$	same
Input	224×224, ImageNet normalization	same
Seeds	42	42, 123, 2026

Table 4. Stage A: lab-trained ResNet-18. Validation is on held-out PLANTVILLAGE images; the last row is the single evaluation on PLANTDOC test images.

Epoch / set	Train acc.	Val./test acc.	Val./test macro-F_1	Val./test loss
1 (lab validation)	0.9502	0.9897	0.9859	0.0375
2 (lab validation)	0.9879	0.9918	0.9897	0.0288
3 (lab validation)	0.9914	**0.9943**	**0.9915**	0.0190
3 (PLANTDOC field test)	–	**0.2544**	**0.2126**	–

5.3 Training recipe and replication

All networks use the same recipe (Table 3): images are resized to 224×224 and normalized with ImageNet statistics; training uses random horizontal flips and $\pm 10^{\circ}$ rotations; the optimizer is AdamW [9] with a constant learning rate. No hyper-parameter was tuned on the test set: the notebook contains no search, and the number of epochs was fixed in advance. Each strategy is run with three seeds $\{42, 123, 2026\}$, which change the seed set, the selection model's head initialization, the random draw, and the final model's initialization and mini-batch order. In total the controlled experiment trains 9 final models and 3 selection models.

6 Results

6.1 RQ1: The lab-to-field gap

Stage A succeeds by the standards of the lab. After one epoch the baseline already reaches 98.97% validation accuracy; after three it reaches **99.43%** accuracy and 0.9915 macro-F_1 (Table 4). Evaluated on the 228 field images, the same network achieves **25.44%** accuracy and 0.2126 macro-F_1 (Figure 3), a drop of 74 accuracy points on classes it has learned perfectly in the lab.

How does the baseline fail?

Figure 4 breaks the field result down by class. Three patterns stand out.

(i) Nine of 26 classes have an F_1 of exactly zero, namely healthy apple, cherry, peach, and strawberry, corn common rust, potato late blight, tomato bacterial spot, tomato leaf mold, and tomato mosaic virus. The best class is healthy grape ($F_1 = 0.74$, with perfect precision but only 58% recall).

(ii) Predictions collapse onto a few classes. Only 21 of the 26 classes are ever predicted. Five classes, namely tomato late blight (28 predictions), tomato early blight (28), healthy soybean (26), bell pepper bacterial spot (22), and corn gray leaf spot (19), absorb 123 of 228 predictions (54%), although no class has

more than 12 true test images (Figure 4b).

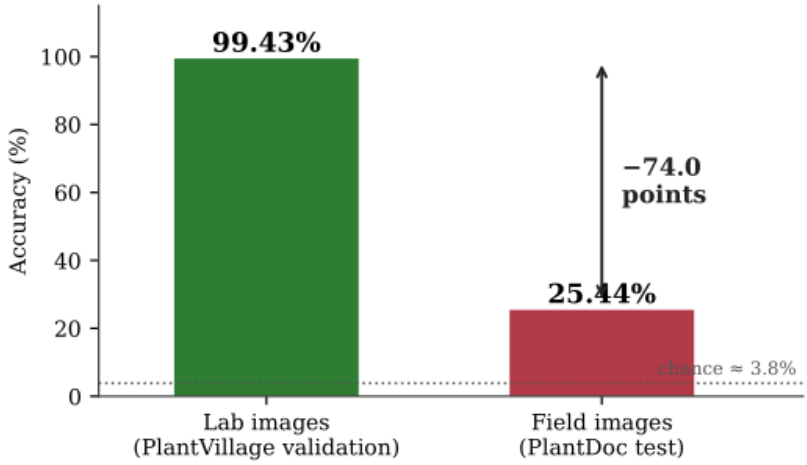


Figure 3. The same network, the same 26 classes: 99.43% on lab images, 25.44% on field images. The dotted line is chance for 26 balanced classes.

(iii) The collapse is not random. A plausible, but untested, explanation is a prior mismatch: the source set is dominated by a few very large classes (healthy soybean has 5,090 images, about 18× the 275 of apple cedar rust) while the field test set is almost balanced. Background shortcuts of the kind documented in PLANTVILLAGE [11] would compound this: a network that learned "plain gray background ⇒ class k" has nothing to rely on once the background changes. Testing these hypotheses would require class-reweighted training and background-ablated inputs, which we leave to future work.

6.2 RQ2: How far do 230 field labels go?

A natural reaction to the gap is to ask for *more lab data*. The experiment suggests that a different kind of data is what is missing. The nine controlled models trained on only 230 field images (0.8% of the lab training set) reach a mean accuracy of 33.1% and a mean macro-F_1 of 0.313, against 25.4% and 0.213 for the lab-trained model. Even the seed set alone, 130 field images (0.46% of the lab data), gave 23.3% accuracy in an exploratory run, nearly matching the lab-trained model with about 217 times fewer images.

Two caveats limit how far this comparison can be pushed. The field-trained models are evaluated in-distribution (the PLANTDOC pool and test set come from the same scraped collection), whereas the baseline is evaluated out-of-distribution, so part of the advantage is simply *being trained on the right kind of picture*. And the baseline is a single run, while the field-trained numbers are averages over nine. The direction of the result is unambiguous; its exact size is not.

6.3 RQ3: Which images should be labeled?

Table 5 lists all nine controlled runs, and Figure 5 summarizes them.

Read literally, Uncertainty wins: it has the highest mean on both metrics (macro-F_1 0.333 versus 0.309 for Random and 0.298 for Diversity), and it wins macro-F_1 on two of three seeds. Random is the most erratic (SD 0.062 for macro-F_1, about twice that of the other strategies): it produced the best single run of the study (seed 42, accuracy 0.399) and also the lowest accuracy (seed 123, 0.294). Diversity never wins macro-F_1 on any seed. This is a consistent *pattern*, but, as the next subsections show, not one that the data can support as a *conclusion*.

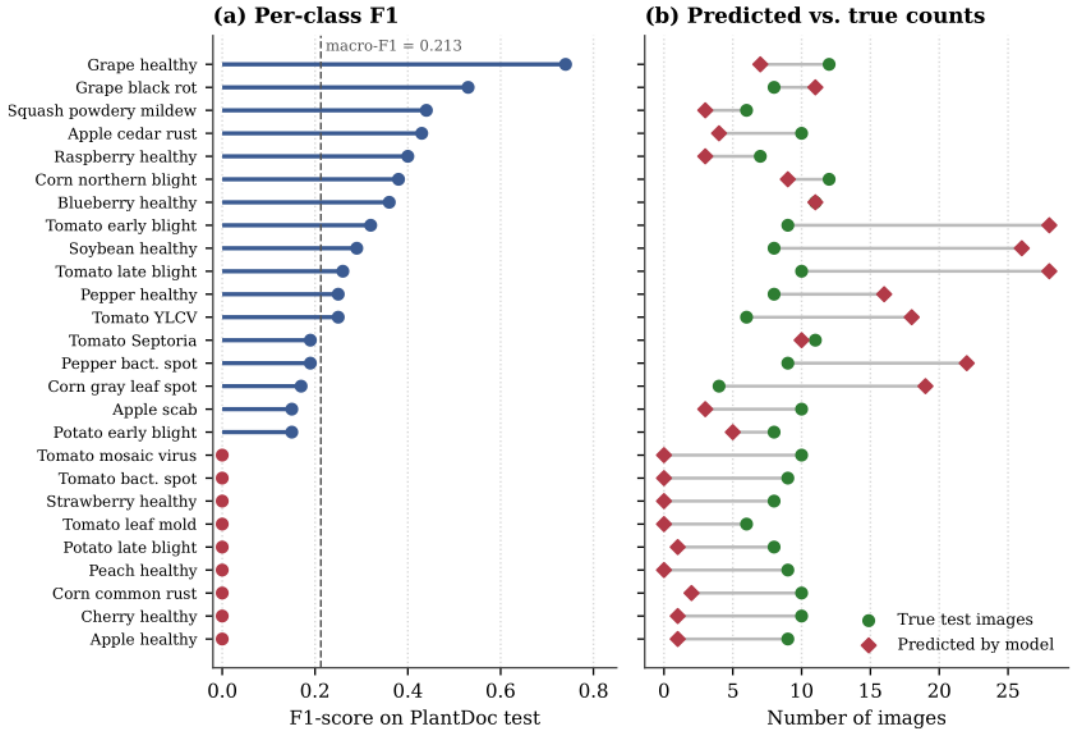


Figure 4. Diagnosis of the lab-trained baseline on the 228 field test images. **(a)** Per-class F_1, sorted; red marks the nine classes with $F_1 = 0$. **(b)** True number of test images per class (green circles) versus the number of images the model assigns to that class (red diamonds): a few classes attract far more predictions than they own, five attract none.

6.4 Graphs produced by the experiment notebook

For completeness, Figure 6 reproduces the dashboard generated directly by the experiment notebook from the nine controlled runs, and Figures 7–10 show the four individual graphs it exported. They display the same numbers as Table 5.

6.5 Ranking instability

The notebook went through three stages, and the apparent winner changed at each one (Table 6). The first run was *exploratory*: strategies were trained one after another in a notebook with shared global state (a module-level optimizer, and a model object reused across stages). In that run Diversity came out on top (accuracy 0.360, macro-F_1 0.330), followed by Uncertainty (0.342, 0.323) and Random (0.320, 0.304). A subsequent audit revealed that the shared state made the comparison unsafe, so the training routine was rewritten to build its own optimizer, all seeds were fixed, cuDNN was made deterministic, and the experiment was rerun. In the controlled seed-42 run the order reversed: Random > Uncertainty > Diversity.

One observation makes the noise floor concrete. The Random strategy's 230 labeled images were, to the best of our reading of the notebook, *identical* in the exploratory run and in the controlled seed-42 run: same seed set, same pool, same random draw. Yet the first model scored **0.320** accuracy and the second **0.399**. The only difference between them is training-time randomness (head initialization, mini-batch order, augmentation draws): a swing of 7.9 points from nondeterminism alone, larger than every between-strategy difference in this study.

Table 5. Controlled active-learning results on the 228 PLANTDOC test images, 230 labels per model. Best value per seed in bold (ties both bold). Bottom block: mean ± sample standard deviation over the three seeds.

Seed	Strategy	Acc.	Macro-F_1	Seed	Strategy	Acc.	Macro-F_1
42	Random	**0.3991**	**0.3801**	2026	Random	0.3026	0.2746
42	Uncertainty	0.3684	0.3598	2026	Uncertainty	**0.3070**	**0.3041**
42	Diversity	0.3509	0.3226	2026	Diversity	**0.3070**	0.2689
123	Random	0.2939	0.2725				
123	Uncertainty	**0.3333**	**0.3341**				
123	Diversity	0.3202	0.3038				

Mean ± SD over seeds			
Random	0.332 ± 0.058	0.309 ± 0.062	highest variance
Uncertainty	**0.336** ± 0.031	**0.333** ± 0.028	best mean on both metrics
Diversity	0.326 ± 0.023	0.298 ± 0.027	lowest mean on both metrics

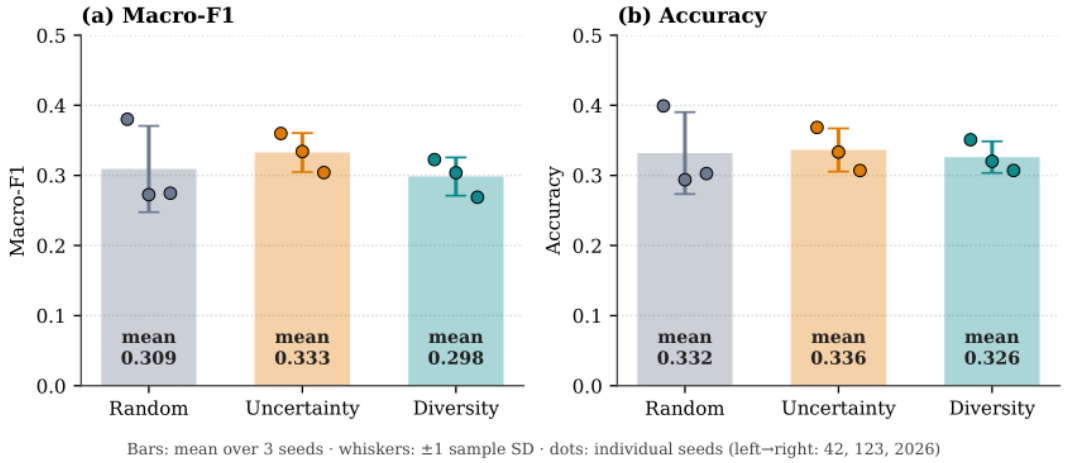


Figure 5. Controlled results over three seeds. Bars show the mean, whiskers one sample standard deviation, dots the individual seeds. The three strategies are not separated by more than the spread of the seeds themselves.

6.6 What did each strategy actually choose?

Although the strategies' outcomes are statistically close, their *behavior* differs clearly. The selection statistics below come from the exploratory run (the only run in which they were tabulated), so they should be read as descriptive.

Overlap. Of 100 images selected by each strategy, only 3 were chosen by both Random and Uncertainty, and 6 by both Random and Diversity, close to the roughly 4.6 expected by chance for two independent draws of 100 from 2,155. Uncertainty and Diversity, however, shared 16 images, suggesting that both are drawn to images a 130-image model finds unfamiliar.

Class coverage. Random touched all 26 classes; Diversity 25; Uncertainty only 21 (Figure 12). Uncertainty sampling asked for *no* images of corn northern blight, corn common rust, tomato yellow leaf curl virus, apple cedar rust, or strawberry, while requesting 14 raspberry and 12 squash powdery mildew images. A natural hypothesis is that the selection model was already (rightly or wrongly) confident on the ignored classes, and entropy cannot see confident mistakes. Diversity sampling over-requested a different set (12 Septoria leaf spot, 9 healthy blueberry) and never asked for potato late blight. This kind of class imbalance in queries is a known risk of purely uncertainty- or distance-based selection, and may be one reason neither strategy dominates.

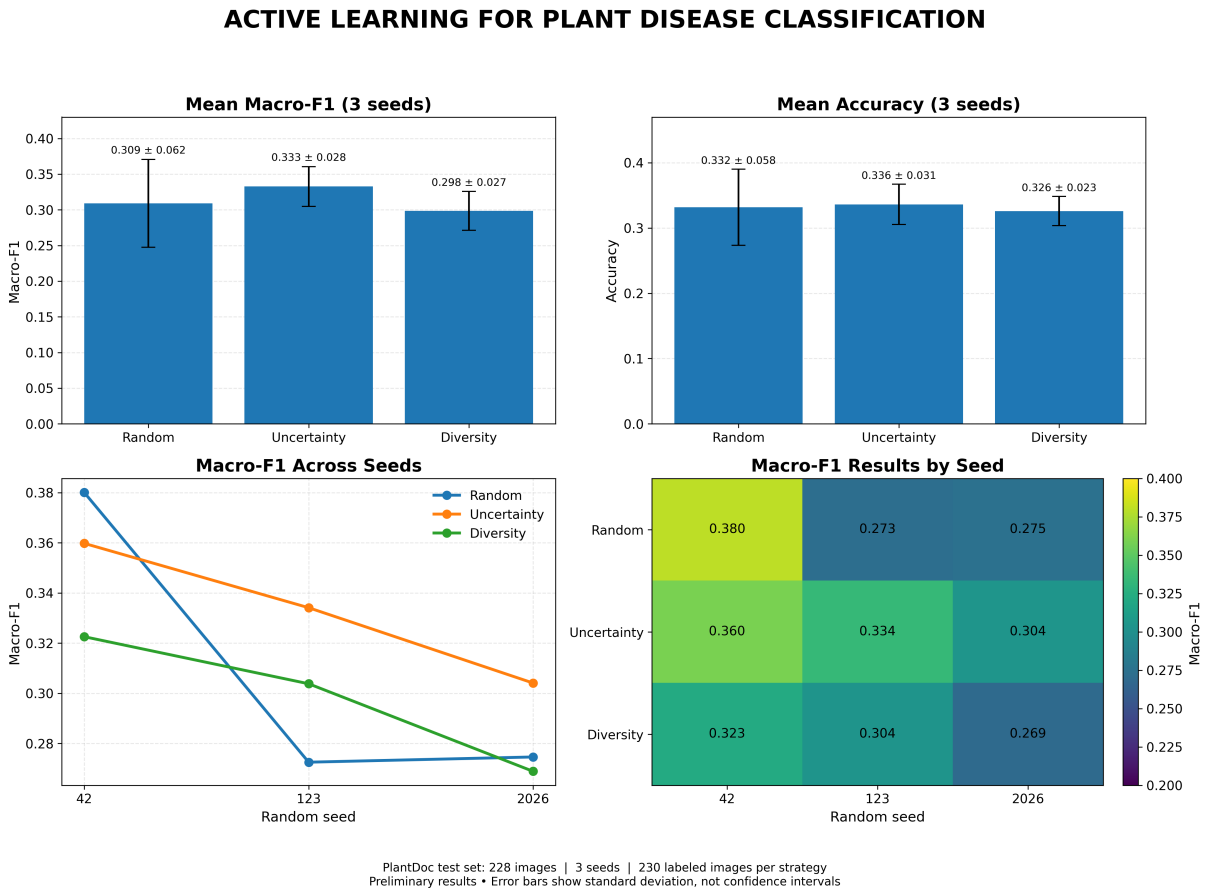


Figure 6. Research dashboard exported by the notebook: mean macro-F_1 and accuracy (error bars are standard deviations), macro-F_1 across seeds, and the per-seed macro-F_1 heatmap.

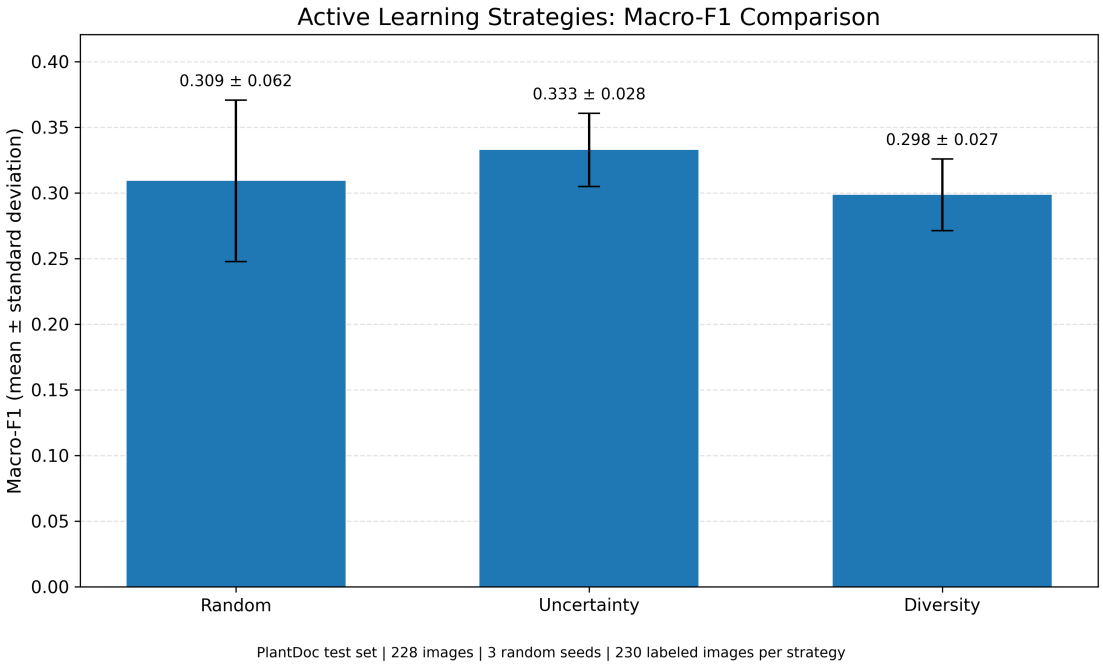


Figure 7. mean macro-F_1 ($\pm$ standard deviation) of the three strategies over three seeds.

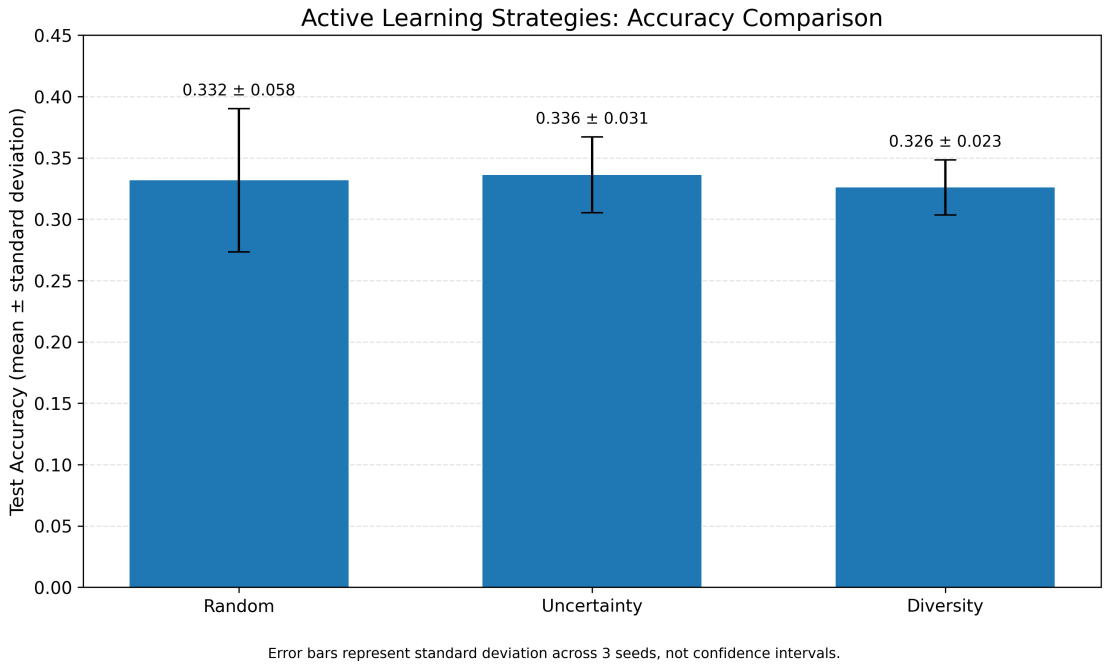


Figure 8. mean test accuracy ($\pm$ standard deviation) of the three strategies over three seeds.

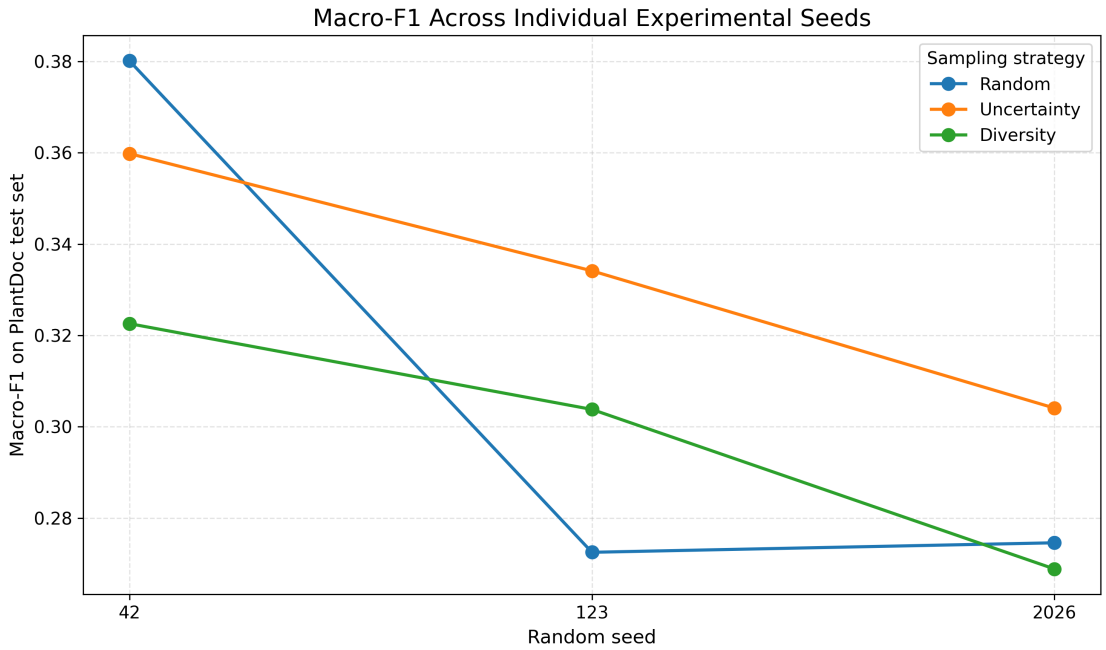


Figure 9. macro-F_1 of each strategy for each individual seed (42, 123, 2026).

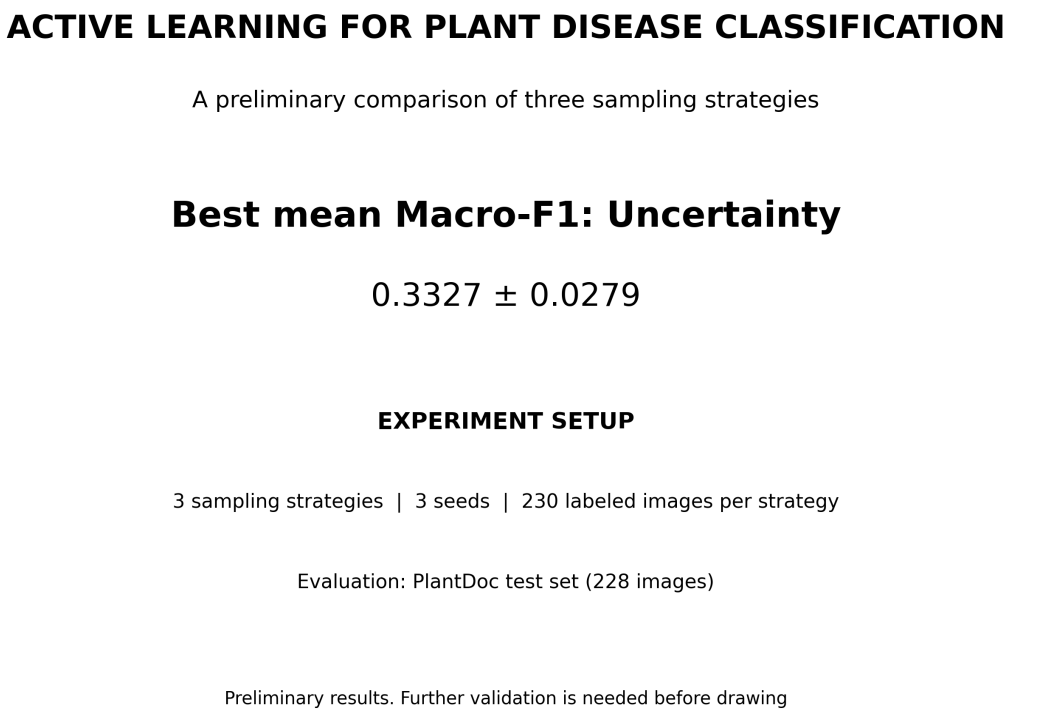


Figure 10. Summary graphic of the experiment.

Table 6. The apparent winner depends on which view of the experiment one reads. The exploratory run is shown for transparency only and is *not* used for any conclusion.

View	Winner (accuracy)	Winner (macro-F_1)	Full order by macro-F_1
Exploratory run (uncontrolled)	Diversity	Diversity	Div. > Unc. > Rand.
Controlled, seed 42	Random	Random	Rand. > Unc. > Div.
Controlled, seed 123	Uncertainty	Uncertainty	Unc. > Div. > Rand.
Controlled, seed 2026	Unc. = Div. (tie)	Uncertainty	Unc. > Rand. > Div.
Mean of three seeds	Uncertainty	Uncertainty	Unc. > Rand. > Div.

6.7 How much noise is there? A statistical reading

Two simple calculations explain why the data cannot separate the strategies.

Test-set sampling error. With $n = 228$ test images and an accuracy near 0.33, the binomial standard error of a single accuracy estimate is $\sqrt{0.33 \times 0.67/228} \approx 0.031$, i.e., about 3.1 points. The three strategies' mean accuracies (0.336, 0.332, 0.326) lie within a 1.0-point range, about one third of that error. Each test image is worth 0.44 points.

The three-seed floor. Paired by seed, Uncertainty beats Random in macro-F_1 by $+0.024$ on average, but with a standard deviation of the paired differences of 0.041 (-0.020, $+0.062$, $+0.030$ for seeds 42, 123, 2026). More fundamentally, an exact two-sided sign-flip test on three pairs has only $2^3 = 8$ equally likely sign patterns, so even a strategy that won on all three seeds could not obtain $p < 2/8 = 0.25$. Three seeds are thus *structurally incapable* of demonstrating significance at conventional levels, a point consistent with the broader recommendation to account for multiple sources of variance in benchmark comparisons [3].

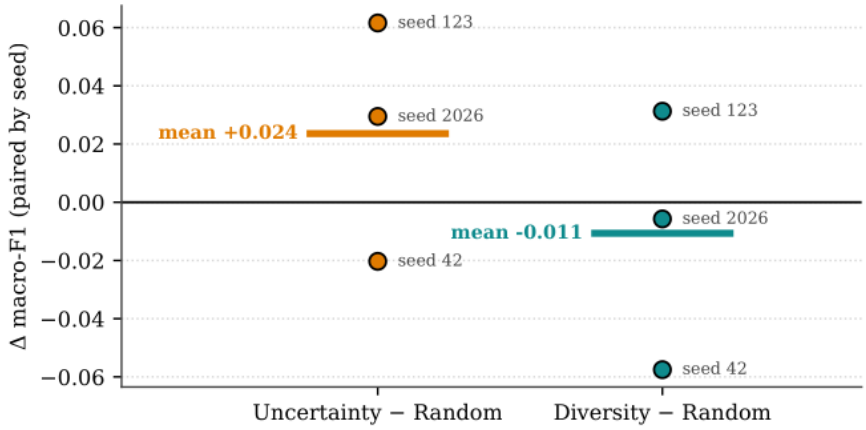


Figure 11. Paired differences in macro-F_1 relative to Random, one dot per seed. Uncertainty beats Random on two seeds and loses on one; Diversity beats Random on one seed and loses on two. Every set of dots straddles zero.

7 Discussion

What the experiment shows. First, the lab-to-field gap is real, large, and specific: a model that is nearly perfect on lab images fails on whole classes of field images. Second, a very small set of in-domain labels recovers a meaningful part of that gap, which reframes the practical question from "how many more images?" to "which images, from where?" Third, at this scale the choice between random, uncertainty, and diversity selection is dominated by noise.

What it does not show. It does *not* show that Uncertainty is better than Random, nor that Diversity is worse. Nor does it show that active learning is useless here: with one round, 100 queries, and three seeds, this design lacks the resolution to detect effects of the 1–3 point size that active-learning methods typically deliver.

Why might Diversity trail? We offer three *hypotheses*, none tested here. (a) Farthest-point selection is attracted to outliers, which in a web-scraped corpus include mislabeled or atypical images. (b) The features ϕ come from a model trained on 130 images for five epochs, so "distance" in that space may not track disease-relevant differences. (c) Both our strategies are of the kind that tend to help at *larger* budgets; the cold-start literature suggests that at very low budgets, selecting typical, representative examples can do better [5].

Practical implications. For a team that has to decide today how to spend a small annotation budget on field images, the evidence in this pilot supports three modest recommendations. (1) Collect *some* in-domain labels before collecting more lab data. (2) Ensure every class is represented in the seed set, since several of the baseline's failures are whole classes. (3) Do not expect a clever acquisition function to outweigh run-to-run noise at this scale; budget for repeated training runs and report spreads rather than single numbers.

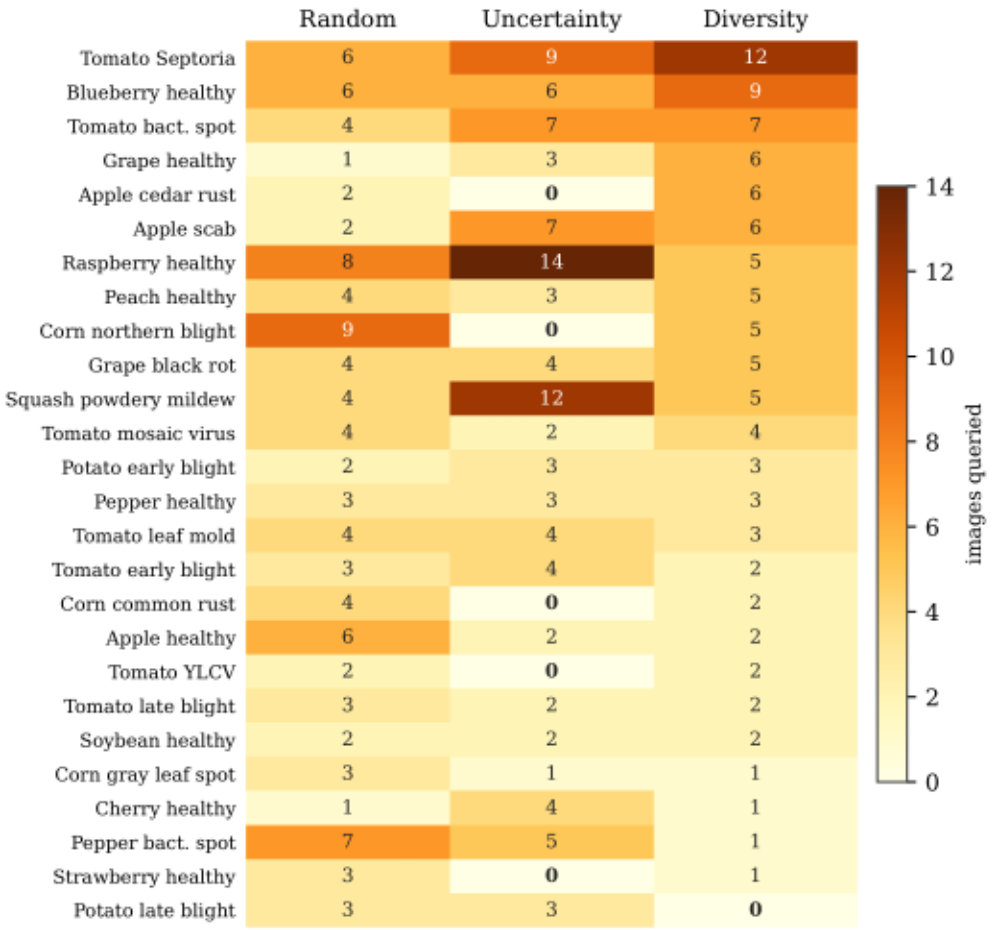


Figure 12. Number of images queried per class by each strategy (100 queries each; exploratory run). Uncertainty leaves five classes unqueried and concentrates on a few; Random is the most even.

8 Limitations and Threats to Validity

- **Small test set.** 228 images across 26 classes (4–12 per class) make every per-class and every between-strategy figure noisy (Section 6.7).
- **Few seeds.** Three seeds cannot support significance tests (Section 6.7). Seed 42 also happens to produce the highest numbers for Random, which pulls its mean upward.
- **One acquisition round.** Real active learning is iterative: retrain, re-score, query again. A single batch of 100 selected from a weak 130-image model is the least favorable setting for uncertainty and feature-based methods.
- **Label harmonization is a judgment call.** Mapping free-form PLANTDOC folders onto PLANTVILLAGE classes (e.g., "Corn leaf blight" → northern leaf blight, "Bell_pepper leaf spot" → bacterial spot) assumes the folder names mean what we think they mean. Any error there is a source of label noise that no strategy can fix.
- **"Field" means web-scraped.** PLANTDOC images are collected from the internet, not captured by farmers in controlled deployment, and the pool and test set share the same collection process. Performance on truly new field photographs may be lower.
- **Leakage audit is exact-match only.** Re-encoded or cropped copies of the same photograph, within PLANTDOC or across the two corpora, would not be detected by SHA-256 digests.

- **Starting point of the active-learning models.** The field-trained models start from ImageNet weights, not from the lab-trained network, so this study does not measure whether lab data *helps* when combined with field labels. The comparison with the baseline in Section 6.1 is therefore indicative, not causal.
- **No tuning.** Hyper-parameters (learning rate, epochs, batch size) were fixed rather than optimized for the very-low-data regime; a tuned recipe might change both absolute numbers and the strategies' ranking.
- **Exploratory selection statistics.** The per-class selection counts and overlaps in Section 6.3 come from the pre-audit exploratory run and are descriptive only.

9 Future Work

1. **Adequately powered comparison.** Repeat the experiment with at least 10–20 seeds per strategy, a larger or cross-validated test set, and paired tests with confidence intervals on the paired differences.
2. **Learning curves.** Replace the single budget with a sweep (for instance 50, 100, 200, 400 queries) and iterative rounds, to find where, if anywhere, the strategies separate from Random.
3. **Combine lab and field data.** Initialize from the lab-trained network, or train jointly with domain-adaptation techniques, to measure how much of the lab data's value survives in the field.
4. **Better selection.** Hybrid strategies that balance uncertainty and diversity, ensemble- or dropout-based uncertainty estimates, and class-balanced query constraints to prevent the 21-class collapse seen for entropy sampling.
5. **Understand the failure.** Class-reweighted training and background-ablation experiments to test the prior-mismatch and shortcut hypotheses of Section 6.1.
6. **Cost-aware annotation.** Not all labels cost the same: distinguishing look-alike diseases takes more expertise than flagging healthy leaves. Acquisition should maximize benefit per unit of expert time, not per image.

10 Conclusion

A plant-disease classifier that is almost perfect in the lab can lose three quarters of its accuracy in the field. The scarce ingredient for repairing it is not compute or lab photographs but expert-labeled field images. In a controlled pilot with a budget of just 100 queries, a few hundred field labels recovered several accuracy points that 28,220 lab images could not, yet the choice of *which* images to label (random, uncertain, or diverse) mattered less than the randomness of training itself. We regard that negative result as informative: it tells us where the next experiment must be stronger, in seeds, in test-set size, and in rounds, before any acquisition function can be credited, or blamed, for what it does.

A Class Mapping

B Per-Class Baseline Results

Table 7. Mapping from PLANTDOC folder names to canonical PLANTVILLAGE classes (26 shared classes). Two PLANTDOC folders ("Tomato leaf" and "Tomato two spotted spider mites leaf") are excluded.

PLANTDOC folder	Canonical (PLANTVILLAGE) class
Apple Scab Leaf	Apple, apple scab
Apple leaf	Apple, healthy
Apple rust leaf	Apple, cedar apple rust
Bell_pepper leaf	Pepper (bell), healthy
Bell_pepper leaf spot	Pepper (bell), bacterial spot
Blueberry leaf	Blueberry, healthy
Cherry leaf	Cherry, healthy
Corn Gray leaf spot	Corn, Cercospora / gray leaf spot
Corn leaf blight	Corn, northern leaf blight
Corn rust leaf	Corn, common rust
Peach leaf	Peach, healthy
Potato leaf early blight	Potato, early blight
Potato leaf late blight	Potato, late blight
Raspberry leaf	Raspberry, healthy
Soyabean leaf	Soybean, healthy
Squash Powdery mildew leaf	Squash, powdery mildew
Strawberry leaf	Strawberry, healthy
Tomato Early blight leaf	Tomato, early blight
Tomato Septoria leaf spot	Tomato, Septoria leaf spot
Tomato leaf bacterial spot	Tomato, bacterial spot
Tomato leaf late blight	Tomato, late blight
Tomato leaf mosaic virus	Tomato, mosaic virus
Tomato leaf yellow virus	Tomato, yellow leaf curl virus
Tomato mold leaf	Tomato, leaf mold
grape leaf	Grape, healthy
grape leaf black rot	Grape, black rot

Table 8. Lab-trained ResNet-18 on the 228 PLANTDOC test images: per-class precision, recall, F_1, and support.

Class	P	R	F_1	n	Class	P	R	F_1	n
Apple scab	.33	.10	.15	10	Raspberry healthy	.67	.29	.40	7
Apple cedar rust	.75	.30	.43	10	Soybean healthy	.19	.62	.29	8
Apple healthy	.00	.00	.00	9	Squash powdery mildew	.67	.33	.44	6
Blueberry healthy	.36	.36	.36	11	Strawberry healthy	.00	.00	.00	8
Cherry healthy	.00	.00	.00	10	Tomato bacterial spot	.00	.00	.00	9
Corn gray leaf spot	.11	.50	.17	4	Tomato early blight	.21	.67	.32	9
Corn common rust	.00	.00	.00	10	Tomato late blight	.18	.50	.26	10
Corn northern blight	.44	.33	.38	12	Tomato leaf mold	.00	.00	.00	6
Grape black rot	.45	.62	.53	8	Tomato Septoria	.20	.18	.19	11
Grape healthy	1.00	.58	.74	12	Tomato YLCV	.17	.50	.25	6
Peach healthy	.00	.00	.00	9	Tomato mosaic virus	.00	.00	.00	10
Pepper bacterial spot	.14	.33	.19	9					
Pepper healthy	.19	.38	.25	8	**Macro average**	.24	.26	**.21**	228
Potato early blight	.20	.12	.15	8	**Accuracy**		**.25**		228
Potato late blight	.00	.00	.00	8					

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